

Practical flood risk reduction strategies in South Sudan

John V. Mayen, MS
Erik Wood, MS
Tim Frazier, PhD

ABSTRACT

More extreme weather patterns caused by climate change are leading to more intense and frequent flooding in some of the world's most vulnerable locations. The Republic of South Sudan faces obstacles mitigating rivers and urban floods caused by heavy rainfall in many of its regions. Through a focused review of the related literature, this study will first analyze the effectiveness and failures of the South Sudanese government's attempts to mitigate flood risks, save lives, and prevent property damage. Study data are derived from various sources including peer-reviewed research, government and nongovernment documents, and the news media. Next, using similar qualitative review methods, a parallel case study from Bangladesh will be used to glean comparative findings and help formulate flood risk reduction recommendations for South Sudan. Findings include that government efforts are highly reactive and lack adequate resources to address flood mitigation. Analysis also indicates that locally driven structural and nonstructural disaster risk reduction efforts are critical to reducing vulnerability to this ongoing and intensifying threat. The study implications and recommendations advance the existing literature on this topic and are relevant to any global region displaying vulnerability opposite the complex threat of repeated intense flooding.

Key words: Flood hazard, climate change, emergency management, disaster risk reduction, South Sudan

INTRODUCTION

Floods are the deadliest natural disasters with frequent occurrences worldwide.^{1,2} Extreme weather

events caused by climate change will bring more frequent and intense flood-related disasters in the next decade.³⁻⁶ Meanwhile, the literature continues to support that low-income status (poverty) is a leading variable of global vulnerability.⁷⁻⁹ "People living in poverty are particularly vulnerable to shocks, including those caused by natural disasters such as floods and droughts."⁴ Floods result from heavy rainfall, hurricanes, storm surge, tsunamis, river overflow, dam failures, overfilled urban drainage basins, and snowmelt. Predicting and mitigating against flooding is challenging when communities or the country at risk has limited resources or expertise in implementing planning, mitigation, recovery, and response strategies.⁴ Mitigating flood hazards requires deploying best practices and structural and nonstructural approaches to strengthen communities' ability to adapt, withstand, and rapidly recover.¹⁰ This study adopts the International Federation of the Red Cross definition of vulnerability as the ability of a person or group of people to cope with, resist, and recover from a hazard, such as repeated flooding.¹¹

Country background

South Sudan is a landlocked country (slightly smaller than Texas) of just over 12 million people—80 percent of which reside in rural areas.¹² It is made up of 10 states and bordered by Sudan, Ethiopia, Kenya, Uganda, the Democratic Republic of Congo, and the Central African Republic.¹³ South Sudan is a relatively new country, established on July 9, 2011, after a long-fought civil war from 1983 to 2005 between North and South which claimed over 2.5 million lives and produced more than 4 million internally

displaced persons (IDPs).^{14,15} The war came to an end with the signing of the Comprehensive Peace Agreement (CPA) supported by the United States and the United Nations.¹⁶ Within South Sudan, the two most powerful warring factions led by President Salva Kiir and Vice President Riek Machar signed a peace treaty that is widely believed to be a fragile accord that could fail.¹⁷ Additionally, during this protracted warring period, the development of critical infrastructure that could mitigate natural disasters, namely, floods, in South Sudan was delayed and remains severely underdeveloped.¹⁵ Mitigating flood risk is vital for supporting community lifelines, continuous critical business, government functions, security, public health, and safety.¹⁸

The climate in South Sudan is tropical—hot all year round, dry in winter, rainy in summer, with less heat and more humidity in winter. South Sudan is vulnerable to several natural hazards, including drought, floods, and climate change variabilities that are likely to negatively impact agriculture due to an increase in the rainfall season.¹⁹ Besides destroying crops, the increase in rainfall can cause waterborne diseases that can rapidly spread among local communities.¹² Ongoing country famine spikes from year to year, while over six million people live with daily food insecurity and 1.1 million children under the age of five are suffering from severe malnutrition.^{20,21} Annual flooding starts in April in South Sudan and renders over 60 percent of the country's inadequate roadways impassable.^{22,23} In fall of 2017, 3,200 pit-dug latrines were flooded lowering sanitation levels and contributing to cholera outbreaks.¹³ This type of flood-related contamination leaves many with no choice but to continue using the same contaminated water for bathing, drinking, and cooking.²⁴ Access to essential services, like healthcare, is nonexistent across much of the country where the ruling government chooses to spend 64 percent of its national budget on weaponizing its military while allocating only 3 percent of the same budget to health care. The result is 1.5 doctors and two nurses for every 100,000 people.²⁵

South Sudan relies heavily on agriculture as the primary source of revenue. The overall poverty rate is high despite large deposits of oil reserves, which

account for 98 percent of the total gross domestic product (GDP).¹² The industry and socio-economic system are severely underdeveloped, while goods and services mostly come from neighboring countries, especially Sudan.¹² With reference to the study area map (Figure 1), 85 percent of the population does nonwage work, eg, subsistence agriculture, and only 1 percent has access to electricity.²² These conditions can produce immediate and severe food security needs along with many compounded threats like gender-based violence (GBV). “During emergencies such as conflicts or natural disasters, the risk of violence, exploitation, and abuse is heightened, particularly for women and girls.”²⁶

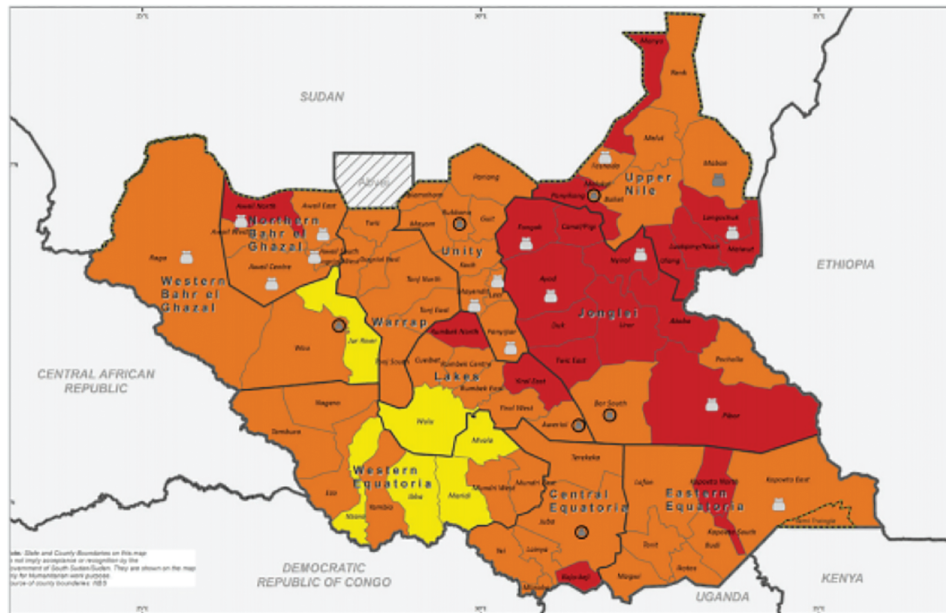
In the United States, local community leaders and decision-makers play a critical role in mitigating risk from hazards by integrating hazard mitigation in their local disaster management plans.²⁸⁻³⁰ Local leaders can set priorities, provide policy outreach, and invite stakeholders to work together to discuss community safety and resilience while effectively communicating mitigation methodologies to large groups of local actors.¹⁸ Conversely, South Sudan's local leaders lack the critical knowledge and ability to mitigate significant disaster outcomes due to an ill-managed governmental system, corruption, tribalism, and resource mismanagement.³¹ This government system relies on assistance from international organizations, eg, United Nations, US Agency for International Development (USAID), and others, to mitigate disaster risk and subsidize basic needs.^{32,33} Within this context, the purpose of this study is to explore and then recommend accessible, locally empowered, and implementable flood risk reduction strategies for South Sudan.

LITERATURE REVIEW

Flooding hazard in South Sudan

The floods in South Sudan have become a recurrent disaster theme, affecting individuals, families, and communities each rainy season for many decades in the past.³⁴ In July 2019, severe flooding devastated many counties and villages across the nation, causing more than 908,000 persons experiencing homelessness (PEH) and forcing the government to declare a

Projected acute food insecurity February - April 2020



Key for the map IPC acute food insecurity Phase classification



Figure 1. Study area with food security overlay (Source: ACAPS²⁷).

state of emergency. The United Nations reported more than 620,000 people as needing immediate, critical life-saving stocks and humanitarian assistance.³⁵ Thirty counties in different states throughout the country were inundated by heavy rainfall and flooding, from Warrap, Northern Bhar el-Gazal, Lake, Upper Nile, Jonglei, Unity to Central and Eastern Equatoria states. The disaster triggered a shortage in essential services, food, medicine, and drinking water for impacted individuals, families, and communities in those 30 counties and beyond.²⁷ The United Nations humanitarian assistance team in South Sudan reported that it needed an estimated \$61.5

million to provide immediate needs to support vulnerable communities and continue disaster operations, response, and recovery.³⁶

South Sudan has a history of flooding, but localised floods have become more frequent. Over 700,000 were affected by severe flooding in the Jonglei and Eastern Nile states in 2020. The floods destroyed crops and displaced almost 85,000 people in Jonglei, in what the World Food Programme (WFP) described as the worst flooding in 60 years.³⁴

The United Nations classified all these counties as a high priority for humanitarian assistance needs.³⁷ In September 2020, for example, river overflow caused by heavy rain flooded communities along the White Nile, which runs in the middle of the country. This disaster has severely impacted several areas, including the Lake Jonglei States. Many people fled their homes to higher ground and could not return home. Simultaneously, conflict displaced more than 33,000 people in this one area alone.²³ Many flood analysts view these flooding events as the worst ever to occur in South Sudan history.^{23,34} The increase in rain in July 2020 caused increased flooding that destroyed farms, washed roads away, led to greater food insecurity, and waterborne disease. This situation has made the recovery process difficult for hundreds of thousands of people during the period of civil war that put the entire country on the brink of collapse and ultimately produced failed state.³⁸ The floods have devastated communities, displaced families, destroyed homes, farms, and crops, and created damages to the environment.³⁸ Flooding is also a driver of systemic malnutrition in South Sudan.³⁹ A climate pattern of less early rains results in crop plants that are small and physically vulnerable when flooding begins. “These conditions prevent people from being able to feed themselves and from building up financial assets.”³⁹ The seasonal, but intensifying, cycle of flooding impacts other, critical “survival sectors” in the South Sudan economy, like the raising of livestock. “Insecurity has brought livestock marketing to a standstill and—the result of poor roads, conflict, and flooding—livestock market integration in South Sudan remains a distant vision.”⁴⁰

Flood risk conceptualization assesses exposure, hazard vulnerability, and consequences and reveals that South Sudan is at high risk from flooding. This calls for immediate and drastic mitigation strategies for managing seasonal flooding.⁴¹ In 2019, the number of people affected by flooding was estimated between 600,000 and 800,000 people in more than seven counties across South Sudan.⁴⁰ Table 1 shows the types of climate change-related disasters and severity levels faced by many areas in South Sudan.

In Figure 2, the flood modeling and analysis of river flooding in South Sudan classified Jonglei,

Table 1. Natural hazards by severity level in South Sudan (Source: Thinkhazard⁴²)

Disaster type	Severity level
Urban flood	High
River flood	High
Wildfire	High
Water scarcity	High
Earthquake	Medium
Volcano	Low
Cyclone	Very low
Coastal flood	No data

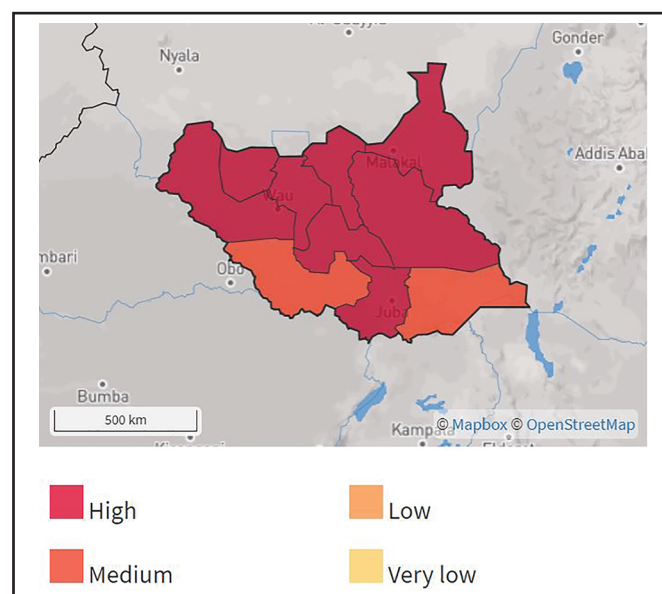


Figure 2. River flood vulnerability for South Sudan (Source: Thinkhazard⁴²).

Unity, Upper Nile, Northern Bahr El Ghazal, Warab, Central Equatoria, and El Buheyrat at high threat levels (red) of river flooding. The flood modeling for the states of Western and Eastern Equatoria (dark orange) indicates medium threat levels.⁴¹ These scenarios required robust project planning decisions and mitigation strategies to prevent the consequences or impacts of river flood hazards in the classified areas. Therefore, a viable solution for reducing community risks associated with river flooding is the implementation of structural mitigation projects.⁴¹

Famine, climate change adaptation, and early warning

The World Bank confirms that South Sudan is vulnerable to several natural hazards, such as drought, floods, and climate change variability.¹² Recent increases in flooding were due to heavy rainfall, and the overflow of water from the Indian Ocean and Lake Victoria—all of which was exacerbated by variabilities in climate change.⁴³ However, there is limited research on the impacts of climate change on migration^{12,44} or the correlation to areas experiencing conflict.⁴⁵ Nevertheless, the United Nations World Food Program USA has indicated that climate change was already impacting South Sudanese communities, reporting that the number of people facing famine in South Sudan and hunger was on the rise, prompting the agency to declare a famine in early 2017. This increase in hunger is due to flooding, drought, and local conflict, impacting an estimated five million people who face food insecurity.⁴⁶ “Extreme food insecurity persists across South Sudan as the lean season progresses, conflict continues to disrupt normal livelihoods, and macroeconomic conditions remain very poor.”⁴⁷

Nongovernmental organizations (NGOs), like the Agency for Technical Cooperation and Development (ACTED), are attempting to mitigate the current food shortages for more than 4.8 million Sudanese people via improved flood resiliency.⁴⁸ “Engaging in specific “adaptation projects” in the most vulnerable areas with stakeholders already active in adaptation, such as community-based organizations, NGO’s and local communities as called for by local leaders, could be an effective measure to address the problems of flooding.”⁴⁹ In 2020, USAID predicted that climate change-related disasters, flooding, and drought in South Sudan were likely to affect populations in the western areas of Jonglei state and Pibor county.⁵⁰ The USAID mission report also indicated that an estimated 8.3 million people (about 70 percent of the population) will need humanitarian aid assistance in 2021 due to the increase in floods, famine, and food insecurity.⁵¹ This prediction reflects an increase in an estimated 800,000 people from the previous fiscal year.

In conjunction with locally driven climate adaptation efforts, improving the hydro-meteorological monitoring network to create an effective and accessible

early warning system (EWS) in South Sudan is a vital objective of this country’s National Adaptation Program of Actions to Climate (NAPA). “Without a functional and efficient early warning system, communities are unprepared and therefore highly vulnerable to floods and droughts.”¹⁹ Indeed, for areas with limited resources, EWS are one of the most cost-effective first steps toward resiliency and adaptation, returning more than four dollars in avoided losses for every dollar invested.⁵² Farmers in these regions see the lack of access to information about climate change and adaptation technologies as one of their most significant barriers to improved resiliency.⁵³ “A focus that neglects to enhance capacity-building and resilience as a prerequisite for managing climate change risks will, in all likelihood, do little to reduce vulnerability to those risks.”⁵⁴ Development of local EWS capabilities requires regional coordination with, for example, the Southern African Regional Climate Outlook Forecast (SARCOF) that monitors current regional hydrological conditions, weather patterns, while also defining current flood risk levels for residents.⁵⁵ “[In South Sudan] Early Warning Systems (EWS) are a key tool that can be utilized to reduce the damage that results from a flood event.”⁴¹

Role of international agencies

The current flooding in South Sudan is a recurring humanitarian crisis, claiming many lives, destroying properties, farms, livestock, and damaging the environment. The United Nations, USAID, and other international organizations operating in the country play an enormous role in emergency support to the local population, providing immediate emergency supplies.³⁴ International organizations working in the country have also provided technical expertise, funding, and advice for the government to engage in disaster planning, preparedness, response, mitigation, and recovery.⁴⁴ As of October 2018, there were approximately 78 international organizations delivering humanitarian service (11 from the United Nations and 67 from other international NGOs) along with 105 local NGOs in the country, minimizing human suffering and saving lives while providing temporary assistance and shelter to all affected.³⁴ The United

Nation's mission in South Sudan reported in 2018 that they needed \$1.5 billion for the entire operation.³⁴

The United Nations Educational, Scientific and Cultural Organization (UNESCO) International Hydrological Program (IHP) has invested heavily in investigating hydrological extremes and climate change in South Sudan. UNESCO and other international organizations have implemented many projects, such as floodplain management and risk management programs.⁵⁶ UNESCO has reputable experience and expertise in floodplain management in countries like Pakistan, Benin, Namibia, and other countries, and it has tried to replicate this in South Sudan.⁵⁵ International banks committed to sustainable development and fair-trade practices can also play a key role in increasing flood resiliency in the mitigation phase through investment and helping to move away from subsistence living.⁵⁷ Next, the United Nations International Strategy for Disaster Reduction (UNISDR) and United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA) see their guiding role on policy guidelines supporting a decentralized flood and drought management effort on the ground that strengthens local communities in the process.⁵⁴ Meanwhile, the United Nations Framework Convention on Climate Change (UNFCCC) helps lower income countries mainstream disaster risk reduction (DRR) efforts with climate change adaptation resulting in national adaptation plans, eg, NAPA, 2015, and program priorities like flood mitigation.⁵⁸ The assistance provided by these international organizations is intended for immediate emergency support and not to replace the whole country's infrastructure system. Additionally, the international organization's aid to assist the country has provided relief to millions of the residents. The South Sudanese government's efforts have not kept up with the crisis and international response. The Department of State attributes this to massive corruption, ill-equipped government institutions, lack of expertise, tribalism, and weak government institutions.⁵⁹

Bangladesh comparative case study

Bangladesh is located in South Asia, shares borders with Myanmar and India, and has a land area of about 147,570 km². The predominantly low-lying

deltaic land was formed by the Brahmaputra, the Ganges, and the Meghna rivers, and more than 310 rivers and tributaries.⁶⁰ These waterways also make the country more vulnerable to several natural disasters, such as cyclones, heavy rainfall, storm surges, riverbank erosion, earthquakes, drought, tsunami, fires, and salinity intrusion. Climate change-related flooding, in particular, impacts the agriculture industry in Bangladesh.⁶¹ Similar to South Sudan, flooding from heavy rainfall and frequent flooding occurs in Bangladesh from as early as May in some areas through at least September leading to loss of human lives, property destruction, and environmental damage.^{59,62} "Bangladesh is widely recognized as one of the most vulnerable countries in the world to extreme climatic events."⁶³

In 2017 alone, flooding forced nine million people into homelessness, and an estimated 1,000 people lost their lives due to drowning and waterborne illness.¹⁴ Bangladesh faced several disasters, such as heavy monsoon rains, cyclone floods in coastal areas, melting waters from the Himalayas, heavy deforestation, expansion of urbanization, and low-lying land loss. Bangladesh is a low-income country with low GDP⁶⁴ and limited resources for flood management and mitigation.⁶⁵ The majority of Bangladesh people cannot afford to pay for flood insurance, making them more vulnerable to flood risks including loss of life and property damage.⁶⁶ While the greatest barrier that remains to the resiliency of Bangladesh today is ongoing and systemic poverty, one of the greatest assets is the changing approach of NGOs who are beginning to embrace disaster risk reduction (DRR) principles found in the Sendai Framework and other international guiding DRR documents.⁶⁴ "Experience from the massive floods in 1987 and 1988, and the cyclone of 1991 has definitely contributed to better tackling of natural disasters today."⁶¹

The government's short-term response efforts include delivering food, water, rescue, shelter, free seed for farmers, and other immediate aid items.⁶⁵ Other long-term response efforts are building embankments, building raised flood shelters, flood warning systems, emergency planning, dams, and reducing deforestation. These long-term measures are not consistently

maintained because they are expensive to manage, and many people see no immediate benefits. However, after being devastated by several disasters, the country decided to shift its response efforts from reactive to a more proactive disaster preparedness model. The National Policy on Disaster Management details “a strategy of preparedness and preventive measures, such as, hazard and risk analysis, land use zoning, building codes and disaster preparedness training; optimum coordination of government and NGO efforts; and inclusion of nonstructural measures such as community disaster preparedness, training and public awareness.”⁶⁵ In Bangladesh, improvements to roads and EWS complement evacuation procedures, and earthen embankments provide a buffer to mitigate storm flooding—but one of the greatest assets is an increase in local disaster planning.⁶⁷ Bangladesh is facing a real threat from several disasters, just like South Sudan and many countries worldwide that are bearing the impacts of climate change.

Bangladesh made progress in poverty reduction by stimulating the economy thereby reducing the number of people living under \$1.90 per day from 44.2 percent to 14.8 percent between 1991 and 2016.⁶⁸ “Post-independence, Bangladesh has been defined by the significant rise in the number of NGOs that have emerged to help the formerly war plagued nation on its way to achieving milestones in economic and infrastructure growth.”⁶⁴ Some initiatives include building 240 cyclone shelters, repairing 387 km of flood embankments, constructing 17,500 hectares of block plantation, building strip plantations in climate change vulnerable areas, and installing 501 solar irrigation pumps.⁶⁷ “The earthen embankments are built to protect the coastal communities from storm surge and tidal/coastal flooding.”⁶⁶ Bangladesh’s government also provided basic adaptation kits, health, water, agricultural goods, and housing to more than 40,000 people residing in saline and floodplain areas. Disaster education and alternative livelihoods support training provided to more than 6,000 households in 200 communities across the country.^{64,67}

The World Bank and the International Development Agency have financed an estimated \$1.5 billion for projects to reduce disaster vulnerability opposite climate

change impacts.⁶⁷ These projects aimed to build coping methods, eg, building schools used as shelters, against cyclones and flooding as in the recent disaster storms that saw a notable reduction in deaths. “Bangladesh has been a frontrunner in adaptation and disaster preparedness.”⁶⁷ Additionally, for more than 40 years, the United Nations and its partners have invested heavily in DRR programs within Bangladesh, helping reduce disaster risks and save lives, properties, and the environment from damages.⁶² Forty years ago, cyclones and flooding resulted in half a million casualties whereas in 2009 the powerful Cyclone Aila claimed a few hundred lives.⁶⁹ The presence of strong leadership sustained international support, and the implementation of structural, eg, EWS, embankments, and nonstructural, eg, community preparedness, projects have resulted in Bangladesh having a much improved national-level preparedness system for DRR.⁶⁸

METHODOLOGY

This research examines the context surrounding the government in South Sudan’s inability to mitigate the recurring hazard of flooding thereby saving lives, preventing property damage, and protecting the environment. Given the distressed economic history of South Sudan, ongoing conflict, and lack of resources, this study is developed around two main queries including:

- Why does the recurring flooding in South Sudan overwhelm the government’s ability to deploy mitigation strategies to reduce flood risk and hazards?
- What DRR strategies can the local, state, and federal governments in South Sudan reasonably deploy to begin to mitigate increasing flood risks, save lives, prevent the loss of property, and protect the environment?

A focused review of the related literature was selected as the methodology. This type of qualitative approach is common as it supports the multidisciplinary nature of emergency management.⁷⁰ Nonquantifiable data are often used in this way to add depth and understanding through legitimate sources

of research.⁷¹ It is also common to employ a short, more precise focused literature review to establish, for example, a novel concept, a policy recommendation, or to encourage additional research.^{72,73} Initially, keyword searches included “South Sudan” + “emergency management” + “flooding hazard” with secondary terms that included “risk reduction” or “conflict” or “humanitarian aid” in varied Boolean arrangements.

Databases accessed through a university library included Emerald, ProQuest Central, EBSCOHost, and JStor. Google Scholar was used to find more related works and to double-check the other databases, while also exploring the core articles,⁴⁸ and where they had been referenced to potentially add more quality citations. The focused review began by reading approximately 71 abstracts and conclusions of potentially related articles. Existing research was deemed even “slightly relevant” or “likely relevant” was read completely for further assessment. Next, 12 core, peer-reviewed articles were retained for being significantly relevant to this topic. From this point, the citation list was expanded in two ways. First, by using the core articles’ references to look backward in time for additional relevant sources and, second, by looking forward at articles that cited those core references. Aside from the peer-reviewed literature, this focused review methodology included data from international organizations operating in the country, news reports, and from government and nongovernment organizations, eg, World Bank, United Nations, US Department of State, and local agencies operating in South Sudan. For the purposes on continuity and study presentation, results of the literature review extended to section 1.0. Similarly, the comparative case study of Bangladesh (2.4) used the same literature review method while only substituting the key word “South Sudan” for the key word “Bangladesh.”

DISCUSSION

This study conducted a focused review of the available literature and a comparable case study review of flood risk reduction efforts in Bangladesh. Results indicate that South Sudan’s government has limited abilities, expertise, policies, willingness, and resources for DRR flood-related projects. When flooding occurs, many South Sudanese people find

themselves reacting to immediate dangers due to the lack of preparedness and access to EWS. In many cases, this leads to panic, catastrophic outcomes, and loss of lives due to the lack of proper disaster preparedness information.⁴² The literature also confirms that there needs to be more of a structural and non-structural commitment to flood mitigation (and DRR in general). Finally, since South Sudan is a relatively young country the international long-term commitment cannot be assessed over the last 40 years, for example, like the case study in Bangladesh. However, it is notable that South Sudan has 78 NGOs operating in the country as of 2018³⁴ while Bangladesh has over 2,500 registered NGOs in the same year.⁶⁴

The critical infrastructure for managing flood risk in South Sudan, ie, drainage systems, canals, urban planning, roads, bridges, dams, building codes, seawalls, and levees, is underdeveloped. The area now known as South Sudan has been in economic distress, fighting civil wars, and experiencing political tension from 1955-1972, 1983-2005, and 2013-2017.²⁷ This history of instability has impeded the development of the types of critical infrastructure that support DRR and meets the UN mandate for sustainable development goals.⁵⁵ These conditions have also created impediments in building capabilities and providing geospatial analysis to help decision-makers at the local and national government levels. The use of geospatial analysis tools, like ArcGIS Online, in the hands of local stakeholders have the potential to judiciously allocate resources to the most vulnerable areas and this is an area ripe for further study in South Sudan. The civil war and many other issues have hindered adaptations to climate change research and resiliency for South Sudan to become sustainable and build disaster risk management capabilities.⁴² These problems lead to weaker governmental institutions, political systems, mismanagement of resources, and delayed economic progress.⁴² Additionally, South Sudan educational institutions have little or no knowledge of flood risk reduction, environmental studies, or emergency management.⁷⁴

Structural approach to disaster risk reduction

Structural mitigation efforts largely apply to physically altering the built environment with, for

example, dams, levees, canals, embankments, or improved drainage systems to reduce the potential loss of life and property damage when water levels rise.⁷⁵ The scale of the project directly relates to a country's economic resources.⁷⁶ In New Orleans, USA, the Seabrook floodgate complex addressed flood risk with a single, permanent \$14.5 billion structural flood mitigation project.⁷⁷ The Seabrook barrier works in conjunction with other existing barriers to reduce storm surge risk to areas, like the Lower Ninth Ward, that were hard hit by Hurricane Katrina in 2005.⁷⁸ Referencing the case study area from Bangladesh, this country has managed to assemble large, phased structural flood mitigation projects starting back in the 1960's.⁷⁹ The current phase has USD \$2 billion budget. "The Government of the People's Republic of Bangladesh is paying for the project—known as Coastal Embankment Improvement Project, Phase-1 (CEIP-1)—with funds borrowed from the World Bank and grant monies issued by the bank under its pilot program for climate resilience."⁷⁷ While these types of projects may not be immediately available in the young country of South Sudan, smaller scale, locally driven structural measures can and must be immediately considered. Flood mitigation products that require more labor, but less direct cost, ie, equipment, specialized tools, and technology, are part of the solution. Two examples of this smaller scale approach to structural mitigation for DRR can include the Trapbag system and Gabion wall systems.

Trapbag systems. TrapBag is a multipurpose system used for flood protection, mudslides, river stabilization, stormwater containment, dams, dikes, levees, seawalls and has a life span of 5 years from the moment of installation.⁸⁰ The TrapBag system (Figure 3), which is cost-effective, easy to ship, and store, protects the entire wall from being compromised if one bag fails as the next one will remain strong, keeping the whole barrier working in the same capacity and providing flood protection.⁷⁸ This system has been used successfully in several US states experiencing the effects of climate change like Illinois and Florida.^{82,83} Referencing Figure 4, several planeloads of these bags were quickly flown to Argentina for



Figure 3. Trapbag example (Source: Sulzberger⁸¹).



Figure 4. Veladero Mine water contamination TrapBag response (Source: Jamasmie⁸⁴).

water containment in response to the 2011 Valadero Gold Mine cyanide spill,^{84,85} which illustrates the potential of this product to be utilized internationally with minimal notice. One key aspect of this system that makes it suitable for South Sudan is that it does not require trained expertise or specific equipment to install. This is the type of local adaptation project that could have an immediate impact on local communities.⁴⁸

Gabion wall systems. Gabion walls are wire mesh boxes that are tied together and filled up with local rocks. This system is less expensive than a concrete wall⁸⁶ and can transition into a more permanent solution than the less expensive TrapBag approach. Gabion walls have the advantage of using local materials and resist water erosion since they freely

drain. However, they are essentially gravity structures that are subject to the lateral force of water.⁸⁷ Including some engineering principles like interlocking the cages instead of stacking them can produce longer term, more effective results.⁸⁵ One such project in Kabul, Afghanistan (Figure 5) protected several villages and valuable agricultural land from flooding. “Under the supervision of HAP [Humanitarian Assistance Program] engineering experts from Kabul, walls were built on both sides of the river to both block and channel floodwater. Local laborers from Kochi Khail were employed during the construction process, helping to benefit the local economy.”⁸⁸ Cesvi, an Italian humanitarian NGO, has also successfully used gabion systems for flood relief in Somalia. A Cesvi employee, working to protect over 600 families from rising river water, explains that “the gabion walls have a stronger resistance to currents, are flexible and adaptable to any changes in the terrain. The simple techniques based on stones locally available make the gabion walls more economically affordable for the communities.”⁸⁹

Another, local flood mitigation project (down to a house-by-house level) employed the gabion wall methodology in Nairobi.⁹⁰ “Localised river remediation and adaptation projects have been carried out at the small scale by governmental and nongovernmental parties in Nairobi.”⁸⁸ However, the degree of expertise, eg, engineering support and oversight, is proportional

to the efficacy of these walls. The gabion wall shown from the project in Nairobi (Figure 6) will likely not outlast the engineered product from Kabul shown in Figure 5, and this may provide insight into why these walls have yielded varying results.⁹¹ In South Sudan, there is a component of urgency where the value of a locally derived effort similar to that shown in Figure 6 would have value as a starting point to flood vulnerability reduction.

Nonstructural approaches to disaster risk reduction

Nonstructural approaches to DRR are less physically obvious than structural ones and relates to what people within a community can offer personally and what personal adoption barriers exist.^{92,93} Nonstructural flood mitigation measures are considered sustainable and universally accepted as being effective.⁹⁴ Some examples are policies, laws, and public education, preparation and evacuation training, early flood warning systems, and forecasting. Enforcement of laws that govern building codes, land use mapping, zoning, and urban planning bring benefits in reducing disaster risk hazards.⁹⁵ Working with local community organizations to disseminate DRR training and education is widely accepted as one of



Figure 5. Gabion wall example from Afghanistan (Source: IOM⁸⁸).

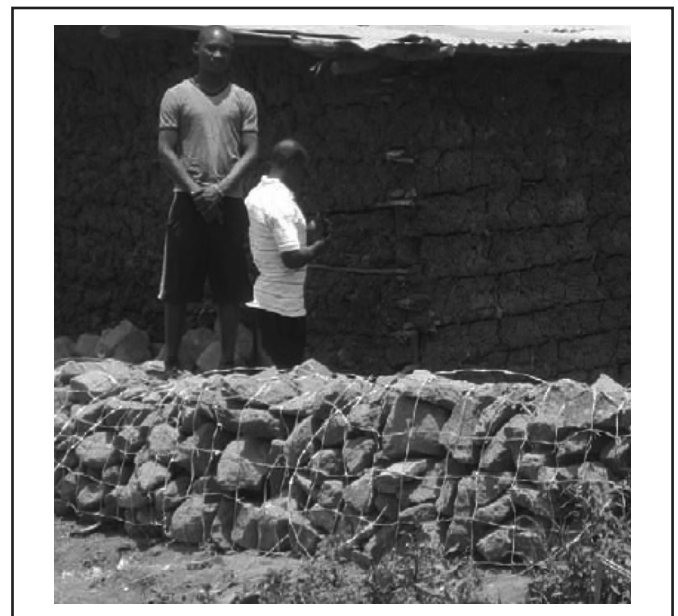


Figure 6. Local village gabion wall in Nairobi (Source: Mulligan et al.⁹⁰).

the best means to raise resiliency. “Emergency planners can better perform their mission if they develop and maintain effective relationships with community- and faith-based organizations in their jurisdictions.”²⁶ This is also true in Bangladesh, where personal resiliency is part of the social fabric. “But risk reduction initiatives don’t necessarily have to involve large-scale construction efforts. Red Crescent volunteers in and around Satiantoli [India] also promote the use of traditional disaster risk reduction related measures, which have been passed down from generation to generation, such as the preparation and storage of clay stoves, the construction of masas, or raised platforms, and the storage of seeds, dry fruit, and grain.”⁹⁶ In fact, several studies have demonstrated the effectiveness of nonstructural flood mitigation approaches in the case study area of Bangladesh.^{97,98} In a lower income setting, like South Sudan, maximizing low-cost, nonstructural mitigation programs may be one of the most accessible means to promote locally empowered DRR.

RECOMMENDATIONS

Based on the findings of this study and the subsequent discussion, five recommendations were derived as a starting point to developing meaningful, practical, and actionable flood risk reduction strategies in South Sudan including:

1. Establish a flood mitigation strategy that brings together local, state, national, and international organizations to work together under one national policy. This includes building partnerships among profit and non-profit, community-based, and faith-based organizations.
2. Develop local DRR through non-structural emergency management training and education programs. This includes strengthening educational institutions to develop a curriculum for teaching natural disaster awareness, flood risk reduction, disaster planning, preparedness, response, mitigation, and recovery.

3. Develop and enhance regionally supported and locally accessible EWS capabilities as part of non-structural flood mitigation measures and connect this to local emergency evacuation training.

4. Implement structural flood mitigation strategies using simple but accessible methods like the Trapbag or Gabion wall systems.

5. Procure funding through international resources like the World Bank to invest in building nationwide critical infrastructure that includes urban planning, land use, hospitals, roads, building codes, bridges, floodwalls, levees, flood barriers, and dams.

CONCLUSION

The ongoing conflict, safety concerns, and travel insecurity prohibited interviewing or gathering other empirical data for this study which may have led to, for example, determining the viability of a low-elevation property buyout program as another means of sustainable flood mitigation in South Sudan. However, this study has produced five actionable and practical recommendations as a meaningful starting point for ongoing flood mitigation efforts for this country that is just beginning only its second decade in existence. Opposite the forces of climate change, flooding in South Sudan will be worse in the next decade. Therefore, a focused effort to use structural and nonstructural means to mitigate this ongoing threat to begin to improve local resilience by moving the people of South Sudan away from subsistence living and toward economic resilience is essential. Additionally, predicting flooding events requires adequate resources and technological innovation which is lacking in countries like South Sudan. The Bangladesh case study and disaster risk reduction programs represent a potential path forward for South Sudan for building their disaster risk reduction strategies. South Sudan has more natural resources than Bangladesh⁹⁹ but having unified voices among government officials to tackle corruption issues can help prevent resource mismanagement, allow the country to invest in sustainable DRR programs, and save lives.

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John V. Mayen, MS, Emergency & Disaster Management, Military Intelligence, Emergency and Disaster Management, Master's Program, School of Continuing Studies, Georgetown University, Washington, DC.

Erik Wood, MS, Adjunct Faculty (Ethics, Research, GIS), Emergency and Disaster Management, Master's Program, School of Continuing Studies, Georgetown University, Washington, DC. ORCID: <https://orcid.org/0000-0003-0531-6314>.

Tim Frazier, PhD, Professor of the Practice and Faculty Director, School of Continuing Studies (SCS), Emergency and Disaster Management, Master's Program, School of Continuing Studies, Georgetown University, Washington, DC. ORCID: <https://orcid.org/0000-0003-1270-5078>.

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